

UNIVERSITY OF PISA
DEPARTMENT OF MATHEMATICS
MASTER'S DEGREE IN MATHEMATICS

Algebraic Number Theory 3

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Academic Year 2026/2027

Last updated: October 23, 2026

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1. p-adic extensions

Definition 1.0.1 (Brauer group)

Let k be a field and k_s be one of its separable closures. The *Brauer group* of k is

$$\mathrm{Br}(k) := H^2(k, k_s)$$

If $K|k$ is a separable subextension of $k_s|k$ we can define the *relative Brauer group* of the extension as

$$\mathrm{Br}(K|k) := \ker\left(\mathrm{Br}(k) \xrightarrow{\mathrm{Res}} \mathrm{Br}(K)\right)$$

The elements of $\mathrm{Br}(k)$ correspond to the finite dimensional division algebras D with center k. In this correspondence, the subgroup $\mathrm{Br}(K|k)$ gives those D for which $D \otimes_k K \simeq M_{n\times n}(K)$, where $n^2 = \dim_k D$.

Lemma 1.0.1

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